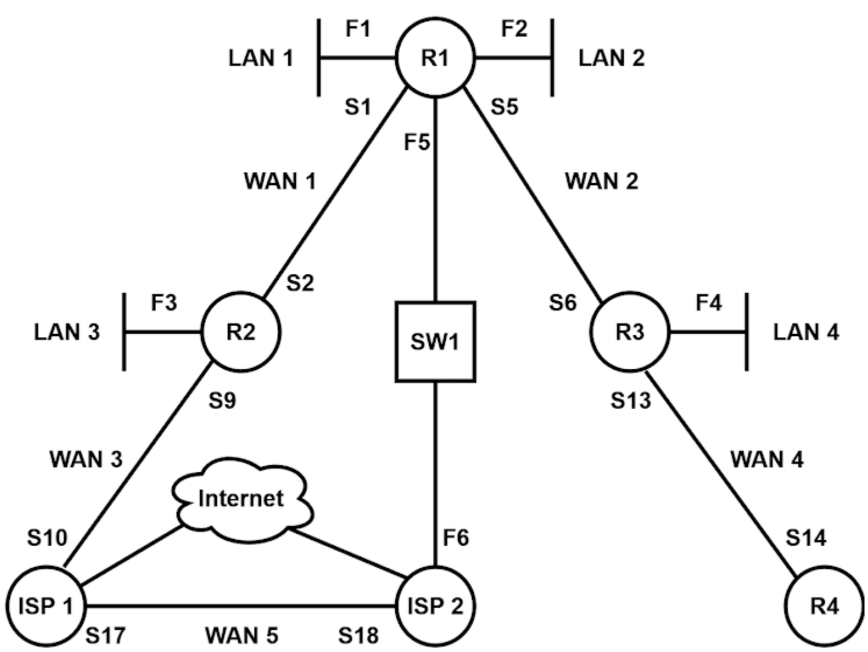
CSE421 Assignment 06 [MSMA | Summer 2025]

Answer all the questions in this form. You can submit only once even if you submit by mistake. Make sure that your answers are put correctly by refreshing the page.

Answer Q1 to Q7 based on the following topology. (If the interface name is F1/S1, the last octet value of that interface's ip address is 1)

Network Addresses of the Networks,

LAN 1 -> 172.16.1.0/25	WAN1 -> 172.16.21.0/30
LAN 2 -> 172.16.2.0/26	WAN2 -> 172.16.21.4/30
LAN 3 -> 172.16.3.0/27	WAN3 -> 172.16.21.8/30
LAN 4 -> 172.16.4.0/28	WAN4 -> 172.16.21.12/30
Switch Network -> 172.16.22.0/29	WAN5 -> 172.16.21.16/30



Q1. In which routers we need to configure default static route?

1. R1
2. R2
3. R3
4. R4
5. ISP1
6. ISP2

Answer: 1, 2, 3, 4

Q2. How many static routes we need to configure for R2? (don't consider internet to be a network)

Answer: 6

If you consider the WAN network between 2 ISPs as part of the internet, then the remote network becomes 6.

Q3. After setting up the IP addresses to all the interfaces of R1, how many C's will be present in R1's routing table?

Answer: 5

Q4. Write the command to create a recursive static route on R2 to reach LAN4. (Use the shortest path)

Answer: ip route 172.16.4.0 255.255.255.240 172.16.21.1

Q5. Configure a directly attached backup route of the above route.

Answer: ip route 172.16.4.0 255.255.255.240 S9 5

Q6. Write the command to create a recursive default static route on R1. (Use the shortest path in terms of hop)

Answer: ip route 0.0.0.0 0.0.0.0 172.16.22.6

Q7. Configure a directly attached backup route of the above default route.

Answer: ip route 0.0.0.0 0.0.0.0 S1 5

Answer Q8 to Q13 based on the following scenario.

```
      192.168.0.0/25 is subnetted, 2 subnets
S        192.168.0.0/25 is directly connected, Serial0/1/0
S        192.168.0.128/25 [1/0] via 192.168.1.241
      192.168.1.0/24 is variably subnetted, 11 subnets, 6 masks
S        192.168.1.0/25 is directly connected, Serial0/1/0
S        192.168.1.128/26 is directly connected, Serial0/1/0
C        192.168.1.192/28 is directly connected, FastEthernet0/0
L        192.168.1.193/32 is directly connected, FastEthernet0/0
C        192.168.1.208/28 is directly connected, FastEthernet0/1
L        192.168.1.209/32 is directly connected, FastEthernet0/1
S        192.168.1.224/29 is directly connected, Serial0/1/0
S        192.168.1.232/29 is directly connected, Serial0/1/0
C        192.168.1.240/30 is directly connected, Serial0/1/0
L        192.168.1.242/32 is directly connected, Serial0/1/0
S        192.168.1.244/30 is directly connected, Serial0/1/0
S*    0.0.0.0/0 [1/0] via 192.168.1.241
```

Q8. Which networks are directly connected to the router?

1. 192.168.0.0/25
2. 192.168.0.128/25
3. 192.168.1.0/25
4. 192.168.1.128/26
5. 192.168.1.192/28
6. 192.168.1.193/32
7. 192.168.1.208/28
8. 192.168.1.209/32
9. 192.168.1.224/29
10. 192.168.1.232/29
11. 192.168.1.240/30
12. 192.168.1.242/32
13. 192.168.1.244/30
14. 0.0.0.0/0

Answer: 5, 7, 11

Q9. Which networks static routing was done using next hop ip address?

1. 192.168.0.0/25
2. 192.168.0.128/25
3. 192.168.1.0/25
4. 192.168.1.128/26
5. 192.168.1.192/28
6. 192.168.1.193/32
7. 192.168.1.208/28
8. 192.168.1.209/32
9. 192.168.1.224/29
10. 192.168.1.232/29
11. 192.168.1.240/30
12. 192.168.1.242/32
13. 192.168.1.244/30
14. 0.0.0.0/0

Answer: 2, 14

Q10. Which networks static routing was done using exit interface?

1. 192.168.0.0/25
2. 192.168.0.128/25
3. 192.168.1.0/25
4. 192.168.1.128/26
5. 192.168.1.192/28
6. 192.168.1.193/32
7. 192.168.1.208/28
8. 192.168.1.209/32
9. 192.168.1.224/29
10. 192.168.1.232/29
11. 192.168.1.240/30
12. 192.168.1.242/32
13. 192.168.1.244/30

14. 0.0.0.0/0

Answer: 1, 3, 4, 9, 10, 13

Q11. Select the IP address of the FastEthernet 0/0 interface?

1. 192.168.0.0/25
2. 192.168.0.128/25
3. 192.168.1.0/25
4. 192.168.1.128/26
5. 192.168.1.192/28
6. 192.168.1.193/32
7. 192.168.1.208/28
8. 192.168.1.209/32
9. 192.168.1.224/29
10. 192.168.1.232/29
11. 192.168.1.240/30
12. 192.168.1.242/32
13. 192.168.1.244/30
14. 0.0.0.0/0

Answer: 6

Q12. What is the value of AD for the 0.0.0.0/0 route?

Answer: 1

Q13. What is the value of cost for the 0.0.0.0/0 route?

Answer: 0